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Rock Piles as Kōura Habitat

Reinforcing kōura resilience using rock piles to enhance freshwater crayfish habitat
in an Aotearoa New Zealand lake

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Placeholder — abstract to be written.

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Freshwater Biology — In preparation

24 Abstract

25 Placeholder — abstract to be written.

26 Introduction

27 Freshwater biodiversity is declining faster than any other biome, with one quarter of freshwater
28 fauna now threatened with extinction (Sayer *et al.*, 2025). Among the most endangered are
29 freshwater crayfish, with 32% of species assessed as threatened globally due to habitat loss,
30 invasive species, and water quality decline (Richman *et al.*, 2015). As keystone species and
31 ecosystem engineers, crayfish play crucial roles in nutrient cycling, bioturbation, and foodweb
1 dynamics, and their loss can have cascading consequences for freshwater ecosystems (Momot,
2 1995; Collier, Parkyn & Rabeni, 1997; Parkyn, Collier & Hicks, 2001).

3 In Aotearoa New Zealand (hereafter Aotearoa), the endemic freshwater crayfish kōura
4 (*Paranephrops planifrons* White, 1842) is a taonga (culturally treasured) species of significant
5 ecological and cultural importance to Māori, particularly for Te Arawa iwi of the Rotorua lakes
6 region in the North Island of Aotearoa (Kusabs & Quinn, 2009). Kōura support customary
7 fisheries using cultural practices such as tau kōura, yet populations in Lakes Rotoiti and
8 Rotorua have declined by up to 96 % over the past two decades, with consequent loss of
9 these customary practices (Kusabs *et al.*, 2015a; Kusabs *et al.*, 2026). The primary driver of
10 decline is predation by the invasive brown bullhead catfish (*Ameiurus nebulosus* Lesueur, 1819,
11 hereafter catfish), confirmed in these lakes in 2016 and 2018 (Francis, 2019; Grayling, 2020).
12 Catfish are native to eastern North America where they mostly live in muddy slow-moving
13 water bodies and can survive and spawn in a wide temperature range, tolerate poor water
14 quality, low oxygen concentration and have a broad diet (Scott & Crossman, 1973), making
15 them a highly adaptable/tolerant invader. Catfish were introduced in Aotearoa in 1877 as
16 sport fish (McDowall, 1994) with their distribution continuously spreading over the upper
17 North Island such as the Waikato River and in 1985 they were detected in Lake Taupō (Barnes
18 & Hicks, 2003). In Aotearoa catfish are a major predator on kōura, as 64 % of large catfish
19 (<250 mm fork length) had kōura in their stomach contents (Barnes & Hicks, 2003). In
20 combination with predation by catfish kōura are further compressed by multiple interacting
21 stressors that progressively restrict their available habitat. Kōura are naturally found in
22 complex rocky habitats (Jowett, Parkyn & Richardson, 2008; Kusabs, Quinn & Hamilton,
23 2015b) or in deeper parts of the lake to avoid predation during the day, migrating to the

productive littoral zone at night to forage (Deveich, 1979). However, thermal stratification creates anoxic conditions in deeper waters, displacing kōura into the shallow littoral zone (Kusabs *et al.*, 2026), precisely where catfish predation pressure is greatest, as catfish rarely swim beyond 17 m depth (Dedual, 2002). This compression is further intensified by dense invasive macrophyte beds obstructing kōura migration in the littoral zone (Kusabs & Quinn, 2009), which also intensify oxygen depletion in deeper waters by die-off events (Vincent, Gibbs & Dryden, 1984; Hamilton, McBride & Uraoka, 2005). The importance of deep refuge is shown by Lake Taupō, where kōura persist at a depth even in the presence of catfish because the lake remains oxygenated below the thermocline (Verburg & Albert, 2019). In Lake Rotoiti this deep refuge becomes anoxic and unavailable for kōura (Kusabs *et al.*, 2026). Next to direct predation catfish exposure triggers anti-predator responses in kōura including increased refuge use (Raven *et al.*, 2026a), reducing time available for foraging and potentially suppressing growth and reproduction (Preisser, Bolnick & Benard, 2005; Sheriff *et al.*, 2020). Climate change is expected to strengthen thermal stratification, expand anoxic zones (Woolway *et al.*, 2021), and increase the range of suitable catfish habitat in Aotearoa, further reducing kōura persistence (Lee *et al.*, 2025).

The compression of kōura into the littoral zone makes the quality of nearshore habitats increasingly important for kōura populations. In stream environments, kōura are positively associated with structurally complex habitats including bank undercuts, woody debris, leaf litter, and coarse substrates (Usio & Townsend, 2000; Parkyn & Collier, 2004; Jowett *et al.*, 2008; Parkyn, Meleason & Davies-Colley, 2009). In lake environments benthic substrate characteristics are a stronger predictor of kōura distribution than nutrient enrichment or predator abundance (Kusabs *et al.*, 2015b), and kōura are positively associated with coarse substrates and riparian vegetation (Raven *et al.*, 2026b). Coarse rocky substrate provides refugia from predators and shelter during moulting (Streissl & Hödl, 2002). Experimental studies confirm that shelter availability directly reduces catfish predation on crayfish (Adams, 2007; Zhao *et al.*, 2024), and kōura increase refuge use in the presence of catfish compared to native predators (Raven *et al.*, 2026a), emphasising the importance of shelter availability for kōura populations. Shelter availability is likely a limiting factor for kōura in Lake Rotoiti, where the majority of the shoreline is sandy with dense beds of introduced macrophytes and little natural rocky structure. Adding rocky substrate to soft-bottomed lake areas has been shown to significantly increase crayfish abundance (Johnsen & Taugbøl, 2008), and increasing structural complexity in streams improves crayfish restocking success (Huolila, Marjomäki & Laukkanen, 1997). Deploying habitat enhancement structures that add structurally complexity in the littoral zone could potentially be a practical way to restore kōura populations in Lake

25 Rotoiti.

26 In this study habitat enhancement structures in the shape of stone piles were deployed along
27 the shoreline of Lake Rotoiti to investigate the ability of the structures in providing additional
28 habitat for kōura to live and hide from predators. Kōura presence, abundance, and biomass
29 were monitored at stone pile, control, and reference sites over 18 months. We predicted that
1 kōura occupancy, abundance and biomass would increase at stone pile sites relative to control
2 sites after construction, and that catfish would not change. We further predicted that kōura
3 abundance would keep increasing progressively after construction, at least within the first 18
4 months, and that catfish would instead show seasonal patterns of shoreline use rather than a
5 construction-driven trend (Barnes, 1996; Dedual, 2002). Finally, we predicted that stone pile
6 sites would progressively converge toward kōura densities observed at natural rocky reference
7 sites, and that catfish would not show this pattern.

8 **Materials and Methods**

9 **Study location and construction**

10 Lake Rotoiti (-38.03, 176.42) is a warm monomictic mesotrophic lake located in the Rotorua Te
11 Arawa Lakes region of the North Island of Aotearoa (34 km², maximum depth 128 m). The lake
12 stratifies for approximately nine months a year. Since 2008, nutrient-rich inflows from Lake
13 Rotorua have been diverted toward the Kaituna River via the Ōhau Channel diversion wall to
14 improve water quality in Lake Rotoiti (Hamilton *et al.*, 2005; Prentice & Özkundakci, 2024).
15 Lake level is regulated by a weir at the outlet of the Kaituna River, providing a stable and
16 controlled water level throughout the year. Catfish are present throughout the lake but occur
17 in particularly high concentrations in Te Weta Bay on the North West side of the lake (Francis,
18 2019). Five stone pile sites were selected alongside the lake shoreline based on accessibility and
19 land owner approval (Figure 1). At each stone pile site, a paired control site was established
20 50 m along the shoreline in the same habitat type. Five naturally occurring rocky shoreline
21 locations were selected as reference sites, each with two monitoring points 50 m apart. In total
22 20 sites of 150 m² (15 * 10 m) were monitored seasonally over 18 months. Baseline monitoring
23 was conducted in February 2025 prior to stone pile deployment, which occurred the following
24 week. Stone piles were constructed from locally sourced quarry rock (Manawahe Stone and
25 Quarry, ~25 km from the lake) with a size range of 50 - 400 mm. At each of the five stone pile
26 sites, three stone piles were arranged horizontally along the shoreline by hand from the shore.

27 Stone pile design was informed by mesocosm experiments demonstrating kōura preference for
28 enclosed stone structures (Raven *et al.*, in review). Stone piles were positioned approximately
29 4.3 m from shore at a mean depth of 0.49 m and with a mean area of 4.1 m².

30 **Monitoring**

31 Monitoring consisted of physical, chemical, and biological parameters recorded at each site
32 during each monitoring round. Static site characteristics including substrate composition, slope
1 to depth contour, riparian vegetation, overhanging tree cover and wood cover were visually
2 estimated at baseline monitoring only, as these were not expected to change substantially over
3 the monitoring period. Substrate composition was visually estimated using an underwater
4 viewing chamber, categorising bedrock, boulders (>256 mm), cobble (64–256 mm), gravel (2–64
5 mm), sand (0.063–2 mm), mud (<0.063 mm), and organic matter. A substrate index was
6 calculated from these estimates (Jowett & Richardson, 1990; Jowett *et al.*, 2008):

$$\text{Substrate index} = 0.08 \times \text{bedrock} + 0.07 \times \text{boulders} + 0.06 \times \text{cobble} + 0.04 \times \text{gravel} + 0.03 \times \text{sand} + 0.02 \times \text{organic matter}$$

7 Coarser substrates were assigned higher weights as they provide greater interstitial pore space
8 and stable structures used by kōura as refuge (Usio & Townsend, 2000; Nyström *et al.*, 2006).
9 After stone pile construction, measurements of the size and dimensions of each structure were
10 taken. The post-construction substrate index for stone pile sites was recalculated using each
11 site's combined stone-pile footprint area, expressed as a percentage of the 150 m² site area
12 and treated as 80 % cobble and 20 % boulders displacing an equivalent area of sand, to reflect
13 their contribution to substrate complexity. Slope was assessed as the horizontal distance from
14 the shoreline to the 20 m depth contour, derived from bathymetric maps and measured in a
15 projected coordinate system (NZTM2000; EPSG:2193).

16 Seasonal variable parameters were recorded at every monitoring round, and included aquatic
17 vegetation cover (emergent and submerged macrophytes) and chemical parameters included
18 water temperature (°C), dissolved oxygen (mg L⁻¹), and conductivity (µS cm⁻¹) were measured
19 using a YSI ProSolo meter (Pro2030, YSI Inc., Yellow Springs, OH, USA). pH was measured
20 using a handheld pH meter (EC-PCTestr35, Eutech Instruments).

21 Kōura and fish were every monitoring round sampled using two single-winged, double-throated
22 fyke nets (600 mm high D-shaped entrance hoop) and two collapsible bait traps (480 × 250 mm;
23 two 50 mm entrance openings) baited with catfish flesh (MacNeil *et al.*, 2025). Fyke nets were

placed perpendicular to the shoreline outside of the stone pile structures or at 10 m intervals, with bait traps positioned between them. All equipment was deployed simultaneously at each site for a 24-h period. Captured kōura were measured for orbital carapace length (OCL) using a digital calliper (0.01 mm precision; 150 mm) and for wet mass using a digital scale (± 1 g; 5 kg capacity, Anko Brand, Kmart Group, Mulgrave, VIC, Australia). Catfish and other fish species, including eels (*Anguilla* spp.), goldfish (*Carassius auratus* (Linnaeus, 1758)), common smelt (*Retropinna retropinna* (Richardson, 1848)), kōaro (*Galaxias brevipinnis* Günther, 1866), rainbow trout (*Oncorhynchus mykiss* Walbaum, 1792), common bully (*Gobiomorphus cotidianus* McDowall & Fulton, 1978), and mosquitofish (*Gambusia affinis* (S. F. Baird & Girard, 1853)), were recorded for abundance and measured for total length, and weighed for wet mass. Catch per unit effort (CPUE) and biomass per unit effort (BPUE) were calculated for all kōura and fish based on the number of nets and traps used at each site for each monitoring round (Kusabs *et al.*, 2015b). Missing kōura weights due to failure of scale ($n = 36$) were estimated using a log $-\log$ length–weight regression fitted to individuals with both measurements ($n = 757$), following standard approaches for weight length relationships in aquatic organisms Figure 5. Predicted values were back-transformed with a lognormal bias correction factor ($10^{\{0.5 \cdot \hat{\sigma}^2\}}$) (Quinn & Keough, 2002).

$$\log_{10}(\text{Weight [g]}) = -2.67 + 2.683 \times \log_{10}(\text{OCL length [mm]})$$

Data analysis

Data analysis was divided into three sections, the first section assessed baseline equivalence between stone pile, control, and reference sites before construction, the second section assessed the impact of stone pile construction and after-construction trajectory on kōura and catfish occupancy, abundance, and biomass, and the third section assessed whether stone pile sites converged toward reference sites and if reference sites remained stable over time. All analysis was conducted in R (version 4.5.2; R Core Team (2025)) using the glmmTMB and emmeans packages (Brooks *et al.*, 2017; Lenth & Piaskowski, 2017).

Site variation before stone pile construction was assessed using paired Wilcoxon signed-rank tests to compare stone pile sites with their paired control sites, and unpaired Wilcoxon rank-sum tests to compare stone pile sites with reference sites. This was done for response variables (water quality, substrate, CPUE, BPUE) to confirm that the stone pile and control sites were equivalent before construction, and to characterise the difference between stone pile and

reference sites. To further assess the baseline equivalence generalised linear models (GLMs) were fitted testing the CPUE and BPUE of kōura and catfish against site type (stone pile/control) and tested each of the two reference sites (reference 1/reference 2) with a Tweedie family and log link. The covariate temperature was included in the catfish abundance and biomass models testing the reference sites to allow model convergence.

The second section of the analysis tested construction impact relative to control sites using a before-after, control-impact (BACI) design ([Underwood, 1994](#)) and tested after-construction trajectory of kōura and catfish populations. The BACI design tested using generalised linear mixed effects models (GLMMs) the effect of stone pile construction on kōura and catfish occupancy, abundance, and biomass by comparing one pre-construction monitoring round with six post-construction monitoring rounds. The models tested period (before/after) and site type (stone pile/control) as an interaction term, with monitoring round as a covariate and site nested within pair as a random effect. Occupancy was tested with a Binomial family and abundance and biomass using a Tweedie family with log link. The after-construction trajectory models were tested using GLMMs with site type (stone pile/control) and monitoring round as interaction terms and site number as random effect using similar families as the BACI models.

The third section of the analysis assessed whether stone pile sites converged toward reference sites and if reference sites remained stable over time. Kōura and catfish abundance, and biomass were tested in GLMMs fitted with site type (stone pile/reference) and monitoring round as interaction terms with site number as random effect using a Tweedie family with log link. The catfish abundance model also included temperature as a covariate. The reference site stability was tested in GLMMs fitted with site type (reference 1/reference 2) and monitoring round as interaction terms with site nested within pair as a random effect also using a Tweedie family with log link. Temperature was included in the catfish abundance and season was included in the kōura abundance as covariates.

Multiple-testing correction using Benjamini–Hochberg correction was applied within each research sub-question across the set of models tested. Robustness of the models was further assessed via sensitivity analyses and all models were tested for uniform distribution, dispersion, and outlier rate using the DHARMA package ([Hartig, 2016](#)). Covariate selection for each model was preformed by comparing Corrected Akaike Information Criterion (AICc) of the base model against models with one additional covariate (temperature, season, or substrate index) added at a time. A covariate was retained only when it reduced AICc by more than 2 units relative to the base model ([Advanced issues and deeper insights, 2002](#)). Temperature was

11 retained in in catfish models where it substantially improved model fit, consistent with catfish
12 being associated with warmer water temperatures ([Scott & Crossman, 1973](#)). For the two
13 reference-site catfish models specifically, the base model failed to converge and temperature
14 was required for numerical stability. Season was retained in the kōura abundance model for
15 reference stability, reflecting the known seasonal variation in kōura abundance ([Kusabs *et*](#)
16 [al., 2015b](#)). Season could not be evaluated for the BACI models as it is collinear with the
17 before/after period contrast.

18 **Results**

19 Figure [2](#)

20 Figure [4](#)

21 *Baseline equivalence*

22 *Construction impact relative to control sites*

23 *Recovery toward reference conditions*

24 Figure [3](#)

1 **Discussion**

2 *Opening*

3 **Baseline and site context***

4 **Construction impact***

5 **Convergence and recovery***

6 **Implications and future directions***

7 Most kōura were found along the edges of the stone piles there for recommended future stone
8 piles should be designed to create the most edges.

9 Stone piles were constructed at the shore line to shallow to reach macrophyte beds which
10 are located further offshore. Future stone pile construction could be extended further offshore
11 to reach the macrophyte beds and test if stone piles could suppress macrophyte growth and
12 create pathways for kōura to move between the deeper parts of the lake and the shore.

1 **Conclusion**

1 **Acknowledgements**

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2 and study the treasured kōura. We are grateful to Soweeta Fort-D'ath and William Anaru
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13 **Authors' contributions**

14 Study conception and design were undertaken by O.V.R., I.A.K.K., R.H., F.J.B. and D.Ö.
15 Fieldwork was executed by O.V.R. Data analysis was performed by O.V.R. I.A.K.K obtained
16 council and iwi consent and provided local ecological knowledge. Manuscript drafting was led
17 by O.V.R. All authors contributed to manuscript revision, approved the results, and the final
18 manuscript.

19 Use of Artificial Intelligence

20 Use of Artificial Intelligence: Claude (Anthropic) was used to assist with manuscript editing and
21 language refinement. All content was critically reviewed and the authors take full responsibility
22 for the final manuscript.

23 Declarations

24 The corresponding author has declared that none of the authors has any competing interests.

25 Data availability statement

1 The code and derived data supporting the findings of this study are openly available on GitHub
2 (https://github.com/OlivierRaven/Stone_Piles_Rotoiti) and a rendered version of the analysis
3 is hosted at https://oliverraven.github.io/Stone_Piles_Rotoiti/. The primary raw dataset is
4 archived on Zenodo (<https://doi.org/.....>). Peer review documents, including the cover letter
5 and reviewer responses, are available in the manuscript repository in the interest of open and
6 transparent science.

7 Tables

Table 1: Baseline (pre-construction) comparability of stone pile, control, and reference sites. Values are mean $\pm$ SD; p-values test Stone pile vs Control (paired Wilcoxon signed-rank test, pairs matched by pair_id, supporting the BACI design) and Stone pile vs Reference (unpaired Wilcoxon rank-sum test, documenting the starting gap relative to natural conditions, motivating the convergence analysis). With only 5 paired sites, these are low-power checks — a non-significant result indicates no obvious gross imbalance, not formal proof of equivalence.

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Kōura CPUE	0.2 $\pm$ 0.27	0.15 $\pm$ 0.22	2.13 $\pm$ 2.32	1.000	0.205
Catfish CPUE	0.05 $\pm$ 0.11	0.1 $\pm$ 0.22	0.3 $\pm$ 0.52	1.000	0.420
Kōura BPUE	7.95 $\pm$ 12.15	2.8 $\pm$ 3.84	36.69 $\pm$ 36.82	0.789	0.206

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Catfish BPUE	0.15 ± 0.34	24.6 ± 55.01	97.28 ± 232.15	1.000	0.343
Substrate index (before)	3.39 ± 1.07	3.68 ± 0.95	5.29 ± 1.05	0.789	0.012
Substrate index (after)	3.65 ± 1.12	3.68 ± 0.95	5.29 ± 1.05	1.000	0.032
Slope (0–20 m)	0.06 ± 0.05	0.06 ± 0.04	0.09 ± 0.05	0.590	0.245
Temperature (°C)	24.74 ± 1.92	24.78 ± 1.47	24.48 ± 0.78	1.000	0.623
Dissolved oxygen (mg/L)	9.85 ± 1.48	9.63 ± 1.29	10.03 ± 0.86	0.418	0.426
Specific conductivity (µS/cm)	195.44 ± 70.05	199.8 ± 78.52	165.45 ± 8.5	0.281	0.951
pH	7.5 ± 2.15	7.61 ± 2.2	8.53 ± 0.66	1.000	0.426

⁸ Source: [Article Notebook](#)

⁹ Figures

¹⁰ Supplementary information

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	re- sponse	com- pari- son	type	time_frame		es- ti- mate	std.er- ror	statis- tic	p.value	adjco- eff	BLow	hig	
M1a	RQ1	Cat- fish	abun- dance	treat- ment	be- fore	(Intercept)	-	0.908	-	0.011	NA	-	-
							2.303	2.535			4.083	0.522	
M1a	RQ1	Cat- fish	abun- dance	treat- ment	be- fore	Site_type_de- tail	-	1.535	-	0.652	0.783	-	2.315
					Stone_pile		0.693	0.452			3.701		

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1	Q2	species	response	com- pari- type	time	frame	es- ti- mate	std.er-	stat-	p.value	adjco-	BFlow	low	high
M1ab	RQ1	Cat-	biomas	treat-	be-	(Intercept)		3.203	1.279	2.504	0.012	NA	0.696	5.709	
		fish		ment	fore										
M1ab	RQ1	Cat-	biomas	treat-	be-	Site_type_de-		-	3.107	-	0.101	0.403	-	0.990	
		fish		ment	fore	tailStone_pile		5.100		1.641			11.190		
M1aa	RQ1	Kōura	abun-	treat-	be-	(Intercept)		-	0.785	-	0.016	NA	-	-	
		dance		ment	fore			1.897		2.415			3.437	0.358	
M1aa	RQ1	Kōura	abun-	treat-	be-	Site_type_de-		0.288	1.044	0.276	0.783	0.783	-	2.334	
		dance		ment	fore	tailStone_pile							1.759		
M1ab	RQ1	Kōura	biomas	treat-	be-	(Intercept)		1.030	0.830	1.241	0.215	NA	-	2.656	
				ment	fore								0.597		
M1ab	RQ1	Kōura	biomas	treat-	be-	Site_type_de-		1.044	0.977	1.068	0.286	0.571	-	2.959	
				ment	fore	tailStone_pile							0.872		
M1ba	RQ1	Cat-	abun-	ref-	be-	(Intercept)		-	32.656	-	0.002	NA	-	-	
		fish	dance	er-	fore			102.299		3.133			166.308	38.295	
				ence											
M1ba	RQ1	Cat-	abun-	ref-	be-	Site_type_de-		-	0.536	-	0.002	0.004	-	-	
		fish	dance	er-	fore	tailReference_2		1.655		3.087			2.706	0.604	
				ence											
M1ba	RQ1	Cat-	abun-	ref-	be-	Temperature		4.079	1.293	3.154	0.002	NA	1.544	6.613	
		fish	dance	er-	fore										
				ence											
M1bb	RQ1	Cat-	biomas	ref-	be-	(Intercept)		-	37.363	-	0.000	NA	-	-	
		fish		er-	fore			153.136		4.099			226.367	79.906	
				ence											
M1bb	RQ1	Cat-	biomas	ref-	be-	Site_type_de-		-	0.743	-	0.001	0.002	-	-	
		fish		er-	fore	tailReference_2		2.569		3.457			4.026	1.112	
				ence											

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1 species	re- response	com- pari- type	time_frame	es- ti- mate	std.er-	stat-	p.value	adjco-	BFlow	low	high
M1bb	RQ1Cat- fish	biomass	ref- er- ence	be- fore	Temperature	6.315	1.482	4.261	0.000	NA	3.410	9.220
M1ba	RQ1Kōura	abun-	ref- er- ence	be- fore	(Intercept)	0.742	0.553	1.343	0.179	NA	-	1.825
		dance									0.341	
M1ba	RQ1Kōura	abun-	ref- er- ence	be- fore	Site_type_de-	0.031	0.778	0.040	0.968	0.968	-	1.557
		dance			tailReference_2						1.494	
M1bb	RQ1Kōura	biomass	ref- er- ence	be- fore	(Intercept)	3.623	0.478	7.580	0.000	NA	2.686	4.560
M1bb	RQ1Kōura	biomass	ref- er- ence	be- fore	Site_type_de-	-	0.681	-	0.952	0.968	-	1.293
		dance			tailReference_2	0.041		0.061			1.375	
M2aa	RQ2Cat- fish	abun-	vs_confull	(Intercept)	-	1.674	-	0.028	NA	-	-	-
		dance	trol			3.679		2.198			6.959	0.398
M2aa	RQ2Cat- fish	abun-	vs_confull	PeriodAfter	0.184	1.470	0.125	0.900	NA	-	3.066	
		dance	trol								2.698	
M2aa	RQ2Cat- fish	abun-	vs_confull	siteStone_pile	-	1.641	-	0.672	NA	-	2.522	
		dance	trol			0.695		0.423			3.912	
M2aa	RQ2Cat- fish	abun-	vs_confull	monitoring_int	-	0.355	-	0.841	NA	-	0.625	
		dance	trol			0.071		0.201			0.768	
M2aa	RQ2Cat- fish	abun-	vs_confull	Peri-	1.742	1.828	0.953	0.341	0.732	-	5.325	
		dance	trol	odAfter:site-							1.842	
				Stone_pile								
M2ab	RQ2Cat- fish	biomass	vs_confull	(Intercept)	1.694	2.773	0.611	0.541	NA	-	7.129	
		trol									3.741	

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q	species	re- response	com- pari- type	time_frame	es- ti- mate	std.er-	stat-	p.value	adjco-	BFlow	low	high
M2ab	RQ26	Cat-	biomass	vs_	confull	PeriodAfter	-	2.260	-	0.709	NA	-	3.587
		fish		trol			0.843	0.373			5.273		
M2ab	RQ26	Cat-	biomass	vs_	confull	siteStone_pile	-	5.456	-	0.341	NA	-	5.500
		fish		trol			5.195	0.952			15.889		
M2ab	RQ26	Cat-	biomass	vs_	confull	monitoring_int	-	0.545	-	0.880	NA	-	0.985
		fish		trol			0.082	0.151			1.150		
M2ab	RQ26	Cat-	biomass	vs_	confull	Peri-	4.977	5.352	0.930	0.352	0.732	-	15.468
		fish		trol		odAfter:site-					5.513		
						Stone_pile							
M2ao	RQ26	Cat-	occu-	vs_	confull	(Intercept)	-	1.963	-	0.281	NA	-	1.730
		fish	pancy	trol			2.117	1.078			5.964		
M2ao	RQ26	Cat-	occu-	vs_	confull	PeriodAfter	-	1.893	-	0.571	NA	-	2.638
		fish	pancy	trol			1.073	0.567			4.784		
M2ao	RQ26	Cat-	occu-	vs_	confull	siteStone_pile	0.000	2.010	0.000	1.000	NA	-	3.939
		fish	pancy	trol							3.939		
M2ao	RQ26	Cat-	occu-	vs_	confull	monitoring_int	0.213	0.331	0.644	0.519	NA	-	0.863
		fish	pancy	trol							0.436		
M2ao	RQ26	Cat-	occu-	vs_	confull	Peri-	0.430	2.216	0.194	0.846	0.846	-	4.774
		fish	pancy	trol		odAfter:site-					3.914		
						Stone_pile							
M2aa	RQ26	Kōura	abun-	vs_	confull	(Intercept)	-	1.178	-	0.020	NA	-	-
		dance	trol				2.739	2.324			5.048	0.429	
M2aa	RQ26	Kōura	abun-	vs_	confull	PeriodAfter	2.121	0.973	2.179	0.029	NA	0.213	4.028
		dance	trol										
M2aa	RQ26	Kōura	abun-	vs_	confull	siteStone_pile	0.714	1.316	0.543	0.587	NA	-	3.293
		dance	trol								1.865		
M2aa	RQ26	Kōura	abun-	vs_	confull	monitoring_int	-	0.093	-	0.189	NA	-	0.061
		dance	trol				0.123	1.312			0.306		

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50	Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60	Q61	Q62	Q63	Q64	Q65	Q66	Q67	Q68	Q69	Q70	Q71	Q72	Q73	Q74	Q75	Q76	Q77	Q78	Q79	Q80	Q81	Q82	Q83	Q84	Q85	Q86	Q87	Q88	Q89	Q90	Q91	Q92	Q93	Q94	Q95	Q96	Q97	Q98	Q99	Q100	Q101	Q102	Q103	Q104	Q105	Q106	Q107	Q108	Q109	Q110	Q111	Q112	Q113	Q114	Q115	Q116	Q117	Q118	Q119	Q120	Q121	Q122	Q123	Q124	Q125	Q126	Q127	Q128	Q129	Q130	Q131	Q132	Q133	Q134	Q135	Q136	Q137	Q138	Q139	Q140	Q141	Q142	Q143	Q144	Q145	Q146	Q147	Q148	Q149	Q150	Q151	Q152	Q153	Q154	Q155	Q156	Q157	Q158	Q159	Q160	Q161	Q162	Q163	Q164	Q165	Q166	Q167	Q168	Q169	Q170	Q171	Q172	Q173	Q174	Q175	Q176	Q177	Q178	Q179	Q180	Q181	Q182	Q183	Q184	Q185	Q186	Q187	Q188	Q189	Q190	Q191	Q192	Q193	Q194	Q195	Q196	Q197	Q198	Q199	Q200	Q201	Q202	Q203	Q204	Q205	Q206	Q207	Q208	Q209	Q210	Q211	Q212	Q213	Q214	Q215	Q216	Q217	Q218	Q219	Q220	Q221	Q222	Q223	Q224	Q225	Q226	Q227	Q228	Q229	Q230	Q231	Q232	Q233	Q234	Q235	Q236	Q237	Q238	Q239	Q240	Q241	Q242	Q243	Q244	Q245	Q246	Q247	Q248	Q249	Q250	Q251	Q252	Q253	Q254	Q255	Q256	Q257	Q258	Q259	Q260	Q261	Q262	Q263	Q264	Q265	Q266	Q267	Q268	Q269	Q270	Q271	Q272	Q273	Q274	Q275	Q276	Q277	Q278	Q279	Q280	Q281	Q282	Q283	Q284	Q285	Q286	Q287	Q288	Q289	Q290	Q291	Q292	Q293	Q294	Q295	Q296	Q297	Q298	Q299	Q300	Q301	Q302	Q303	Q304	Q305	Q306	Q307	Q308	Q309	Q310	Q311	Q312	Q313	Q314	Q315	Q316	Q317	Q318	Q319	Q320	Q321	Q322	Q323	Q324	Q325	Q326	Q327	Q328	Q329	Q330	Q331	Q332	Q333	Q334	Q335	Q336	Q337	Q338	Q339	Q340	Q341	Q342	Q343	Q344	Q345	Q346	Q347	Q348	Q349	Q350	Q351	Q352	Q353	Q354	Q355	Q356	Q357	Q358	Q359	Q360	Q361	Q362	Q363	Q364	Q365	Q366	Q367	Q368	Q369	Q370	Q371	Q372	Q373	Q374	Q375	Q376	Q377	Q378	Q379	Q380	Q381	Q382	Q383	Q384	Q385	Q386	Q387	Q388	Q389	Q390	Q391	Q392	Q393	Q394	Q395	Q396	Q397	Q398	Q399	Q400	Q401	Q402	Q403	Q404	Q405	Q406	Q407	Q408	Q409	Q410	Q411	Q412	Q413	Q414	Q415	Q416	Q417	Q418	Q419	Q420	Q421	Q422	Q423	Q424	Q425	Q426	Q427	Q428	Q429	Q430	Q431	Q432	Q433	Q434	Q435	Q436	Q437	Q438	Q439	Q440	Q441	Q442	Q443	Q444	Q445	Q446	Q447	Q448	Q449	Q450	Q451	Q452	Q453	Q454	Q455	Q456	Q457	Q458	Q459	Q460	Q461	Q462	Q463	Q464	Q465	Q466	Q467	Q468	Q469	Q470	Q471	Q472	Q473	Q474	Q475	Q476	Q477	Q478	Q479	Q480	Q481	Q482	Q483	Q484	Q485	Q486	Q487	Q488	Q489	Q490	Q491	Q492	Q493	Q494	Q495	Q496	Q497	Q498	Q499	Q500	Q501	Q502	Q503	Q504	Q505	Q506	Q507	Q508	Q509	Q510	Q511	Q512	Q513	Q514	Q515	Q516	Q517	Q518	Q519	Q520	Q521	Q522	Q523	Q524	Q525	Q526	Q527	Q528	Q529	Q530	Q531	Q532	Q533	Q534	Q535	Q536	Q537	Q538	Q539	Q540	Q541	Q542	Q543	Q544	Q545	Q546	Q547	Q548	Q549	Q550	Q551	Q552	Q553	Q554	Q555	Q556	Q557	Q558	Q559	Q560	Q561	Q562	Q563	Q564	Q565	Q566	Q567	Q568	Q569	Q570	Q571	Q572	Q573	Q574	Q575	Q576	Q577	Q578	Q579	Q580	Q581	Q582	Q583	Q584	Q585	Q586	Q587	Q588	Q589	Q590	Q591	Q592	Q593	Q594	Q595	Q596	Q597	Q598	Q599	Q600	Q601	Q602	Q603	Q604	Q605	Q606	Q607	Q608	Q609	Q610	Q611	Q612	Q613	Q614	Q615	Q616	Q617	Q618	Q619	Q620	Q621	Q622	Q623	Q624	Q625	Q626	Q627	Q628	Q629	Q630	Q631	Q632	Q633	Q634	Q635	Q636	Q637	Q638	Q639	Q640	Q641	Q642	Q643	Q644	Q645	Q646	Q647	Q648	Q649	Q650	Q651	Q652	Q653	Q654	Q655	Q656	Q657	Q658	Q659	Q660	Q661	Q662	Q663	Q664	Q665	Q666	Q667	Q668	Q669	Q670	Q671	Q672	Q673	Q674	Q675	Q676	Q677	Q678	Q679	Q680	Q681	Q682	Q683	Q684	Q685	Q686	Q687	Q688	Q689	Q690	Q691	Q692	Q693	Q694	Q695	Q696	Q697	Q698	Q699	Q700	Q701	Q702	Q703	Q704	Q705	Q706	Q707	Q708	Q709	Q710	Q711	Q712	Q713	Q714	Q715	Q716	Q717	Q718	Q719	Q720	Q721	Q722	Q723	Q724	Q725	Q726	Q727	Q728	Q729	Q730	Q731	Q732	Q733	Q734	Q735	Q736	Q737	Q738	Q739	Q740	Q741	Q742	Q743	Q744	Q745	Q746	Q747	Q748	Q749	Q750	Q751	Q752	Q753	Q754	Q755	Q756	Q757	Q758	Q759	Q760	Q761	Q762	Q763	Q764	Q765	Q766	Q767	Q768	Q769	Q770	Q771	Q772	Q773	Q774	Q775	Q776	Q777	Q778	Q779	Q780	Q781	Q782	Q783	Q784	Q785	Q786	Q787	Q788	Q789	Q790	Q791	Q792	Q793	Q794	Q795	Q796	Q797	Q798	Q799	Q800	Q801	Q802	Q803	Q804	Q805	Q806	Q807	Q808	Q809	Q810	Q811	Q812	Q813	Q814	Q815	Q816	Q817	Q818	Q819	Q820	Q821	Q822	Q823	Q824	Q825	Q826	Q827	Q828	Q829	Q830	Q831	Q832	Q833	Q834	Q835	Q836	Q837	Q838	Q839	Q840	Q841	Q842	Q843	Q844	Q845	Q846	Q847	Q848	Q849	Q850	Q851	Q852	Q853	Q854	Q855	Q856	Q857	Q858	Q859	Q860	Q861	Q862	Q863	Q864	Q865	Q866	Q867	Q868	Q869	Q870	Q871	Q872	Q873	Q874	Q875	Q876	Q877	Q878	Q879	Q880	Q881	Q882	Q883	Q884	Q885	Q886	Q887	Q888	Q889	Q890	Q891	Q892	Q893	Q894	Q895	Q896	Q897	Q898	Q899	Q900	Q901	Q902	Q903	Q904	Q905	Q906	Q907	Q908	Q909	Q910	Q911	Q912	Q913	Q914	Q915	Q916	Q917	Q918	Q919	Q920	Q921	Q922	Q923	Q924	Q925	Q926	Q927	Q928	Q929	Q930	Q931	Q932	Q933	Q934	Q935	Q936	Q937	Q938	Q939	Q940	Q941	Q942	Q943	Q944	Q945	Q946	Q947	Q948	Q949	Q950	Q951	Q952	Q953	Q954	Q955	Q956	Q957	Q958	Q959	Q960	Q961	Q962	Q963	Q964	Q965	Q966	Q967	Q968	Q969	Q970	Q971	Q972	Q973	Q974	Q975	Q976	Q977	Q978	Q979	Q980	Q981	Q982	Q983	Q984	Q985	Q986	Q987	Q988	Q989	Q990	Q991	Q992	Q993	Q994	Q995	Q996	Q997	Q998	Q999	Q1000
re-	com-	pari-	time	frame	es-	std.er-	stat-	p.value	adjco	BFlow	conf.high																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
M2aa	RQ2K	abun-	vs_conf	full	Peri-	0.286	1.224	0.234	0.815	0.846	- 2.685																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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M2ab	RQ2K	biomass	vs_conf	full	(Intercept)	0.353	1.341	0.263	0.793	NA	- 2.980																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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M2ab	RQ2K	biomass	vs_conf	full	PeriodAfter	1.982	1.259	1.574	0.115	NA	- 4.451																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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M2ab	RQ2K	biomass	vs_conf	full	siteStone_pile	1.370	1.542	0.888	0.375	NA	- 4.393																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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M2ab	RQ2K	biomass	vs_conf	full	monitoring_int	- 0.126	- 0.656	NA	- 0.191																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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M2ab	RQ2K	biomass	vs_conf	full	Peri-	- 1.477	- 0.778	0.846	- 2.479																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
		trol			odAfter:site-	0.416	0.282		3.311																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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M2ao	RQ2K	occu-	vs_conf	full	(Intercept)	- 1.727	- 0.687	NA	- 2.689																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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M2ao	RQ2K	occu-	vs_conf	full	PeriodAfter	1.019	1.571	0.648	0.517	NA	- 4.098																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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M2ao	RQ2K	occu-	vs_conf	full	siteStone_pile	- 2.277	- 0.988	NA	- 4.429																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
		pancy	trol			0.035	0.015		4.498																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
M2ao	RQ2K	occu-	vs_conf	full	monitoring_int	0.000	0.287	0.000	1.000	NA	- 0.563																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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M2ao	RQ2K	occu-	vs_conf	full	Peri-	1.711	1.892	0.904	0.366	0.732	- 5.420																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
		pancy	trol		odAfter:site-						1.998																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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M2ba	RQ2K	abun-	vs_conf	af-	(Intercept)	- 1.699	- 0.236	NA	- 1.315																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
		fish	dance	trol		2.016	1.186		5.346																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q	species	response	com- re- pari- type	time_frame	es- ti- std.er-stat- mate ror tic p.value	adjcoBFlow	low	high
M2ba	RQ26	Cat-	abun-	vs_conaf-	siteStone_pile	- 2.471	- 0.274	NA	- 2.142
		fish	dance	trol ter		2.702	1.093	7.546	
M2ba	RQ26	Cat-	abun-	vs_conaf-	monitoring_int	- 0.474	- 0.218	NA	- 0.345
		fish	dance	trol ter		0.585	1.232	1.515	
M2ba	RQ26	Cat-	abun-	vs_conaf-	site-	1.171	0.611	1.917	0.055
		fish	dance	trol ter	Stone_pile:mon-				0.121
					itoring_int				0.026
M2bb	RQ26	Cat-	biomass	vs_conaf-	(Intercept)	3.626	1.976	1.835	0.066
		fish		trol ter					NA
									0.247
M2bb	RQ26	Cat-	biomass	vs_conaf-	siteStone_pile	- 4.616	- 0.162	NA	- 2.600
		fish		trol ter		6.448	1.397	15.496	
M2bb	RQ26	Cat-	biomass	vs_conaf-	monitoring_int	- 0.681	- 0.207	NA	- 0.474
		fish		trol ter		0.860	1.263	2.194	
M2bb	RQ26	Cat-	biomass	vs_conaf-	site-	1.925	1.211	1.589	0.112
		fish		trol ter	Stone_pile:mon-				0.121
					itoring_int				0.449
M2bo	RQ26	Cat-	occu-	vs_conaf-	(Intercept)	- 1.837	- 0.551	NA	- 2.505
		fish	pancy	trol ter		1.095	0.596	4.695	
M2bo	RQ26	Cat-	occu-	vs_conaf-	siteStone_pile	- 2.879	- 0.290	NA	- 2.594
		fish	pancy	trol ter		3.049	1.059	8.692	
M2bo	RQ26	Cat-	occu-	vs_conaf-	monitoring_int	- 0.487	- 0.389	NA	- 0.534
		fish	pancy	trol ter		0.419	0.862	1.373	
M2bo	RQ26	Cat-	occu-	vs_conaf-	site-	1.128	0.701	1.610	0.107
		fish	pancy	trol ter	Stone_pile:mon-				0.121
					itoring_int				0.245
M2ba	RQ26	Kōura	abun-	vs_conaf-	(Intercept)	- 0.911	- 0.886	NA	- 1.655
			dance	trol ter		0.130	0.143	1.915	

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50	Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60	Q61	Q62	Q63	Q64	Q65	Q66	Q67	Q68	Q69	Q70	Q71	Q72	Q73	Q74	Q75	Q76	Q77	Q78	Q79	Q80	Q81	Q82	Q83	Q84	Q85	Q86	Q87	Q88	Q89	Q90	Q91	Q92	Q93	Q94	Q95	Q96	Q97	Q98	Q99	Q100
re-	com-	pari-	time	frame	es-	ti-	std.er-	stat-	p.value	adjco	BFlow	low	high																																																																																							
M2ba	RQ2K	ōura	abun-	vs_conaf-	site	Stone_pile	0.177	1.231	0.144	0.886	NA	-	2.589																																																																																							
		dance	trol	ter								2.235																																																																																								
M2ba	RQ2K	ōura	abun-	vs_conaf-	monitoring_int		-	0.164	-	0.044	NA	-	-																																																																																							
		dance	trol	ter			0.329		2.014			0.650	0.009																																																																																							
M2ba	RQ2K	ōura	abun-	vs_conaf-	site-		0.318	0.205	1.552	0.121	0.121	-	0.718																																																																																							
		dance	trol	ter	Stone_pile:mon-							0.083																																																																																								
					itoring_int																																																																																															
M2bb	RQ2K	ōura	biomass	vs_conaf-	(Intercept)		3.105	0.891	3.487	0.000	NA	1.360	4.850																																																																																							
			trol	ter																																																																																																
M2bb	RQ2K	ōura	biomass	vs_conaf-	site	Stone_pile	-	1.190	-	0.821	NA	-	2.062																																																																																							
			trol	ter			0.270		0.227			2.601																																																																																								
M2bb	RQ2K	ōura	biomass	vs_conaf-	monitoring_int		-	0.220	-	0.124	NA	-	0.093																																																																																							
			trol	ter			0.339		1.538			0.771																																																																																								
M2bb	RQ2K	ōura	biomass	vs_conaf-	site-		0.437	0.276	1.584	0.113	0.121	-	0.977																																																																																							
			trol	ter	Stone_pile:mon-							0.104																																																																																								
					itoring_int																																																																																															
M2bo	RQ2K	ōura	æccu-	vs_conaf-	(Intercept)		1.529	2.376	0.643	0.520	NA	-	6.185																																																																																							
		pancy	trol	ter								3.128																																																																																								
M2bo	RQ2K	ōura	æccu-	vs_conaf-	site	Stone_pile	-	3.580	-	0.686	NA	-	5.571																																																																																							
		pancy	trol	ter			1.447		0.404			8.465																																																																																								
M2bo	RQ2K	ōura	æccu-	vs_conaf-	monitoring_int		-	0.398	-	0.287	NA	-	0.356																																																																																							
		pancy	trol	ter			0.424		1.066			1.203																																																																																								
M2bo	RQ2K	ōura	æccu-	vs_conaf-	site-		1.325	0.827	1.602	0.109	0.121	-	2.946																																																																																							
		pancy	trol	ter	Stone_pile:mon-							0.296																																																																																								
					itoring_int																																																																																															
M3aa	RQ3C	at-	abun-	vs_ref-af-	(Intercept)		-	1.347	-	0.000	NA	-	-																																																																																							
		fish	dance	er-			6.558		4.869			9.198	3.918																																																																																							
				ence																																																																																																

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q	species	re- response	com- pari- type	time_frame	es- ti- mate	std.er- ror	stat- tic	p.value	adjco- B	Flow	low	high
M3aadRQ36	Cat	abun-	vs_ref-	af-	siteStone_pile	-	2.048	-	0.185	NA	-	1.298	
	fish	dance	er-	ter		2.717		1.326			6.732		
			ence										
M3aadRQ36	Cat	abun-	vs_ref-	af-	monitoring_int	-	0.158	-	0.364	NA	-	0.166	
	fish	dance	er-	ter		0.143		0.907			0.453		
			ence										
M3aadRQ36	Cat	abun-	vs_ref-	af-	Temperature	0.273	0.065	4.183	0.000	NA	0.145	0.401	
	fish	dance	er-	ter									
			ence										
M3aadRQ36	Cat	abun-	vs_ref-	af-	site-	0.579	0.434	1.334	0.182	0.243	-	1.429	
	fish	dance	er-	ter	Stone_pile:mon-						0.271		
			ence		itoring_int								
M3abRQ36	Cat	biomass	vs_ref-	af-	(Intercept)	2.334	1.213	1.924	0.054	NA	-	4.711	
	fish		er-	ter							0.044		
			ence										
M3abRQ36	Cat	biomass	vs_ref-	af-	siteStone_pile	-	4.117	-	0.102	NA	-	1.333	
	fish		er-	ter		6.736		1.636			14.806		
			ence										
M3abRQ36	Cat	biomass	vs_ref-	af-	monitoring_int	-	0.222	-	0.919	NA	-	0.413	
	fish		er-	ter		0.023		0.102			0.458		
			ence										
M3abRQ36	Cat	biomass	vs_ref-	af-	site-	1.300	0.923	1.407	0.159	0.243	-	3.109	
	fish		er-	ter	Stone_pile:mon-						0.510		
			ence		itoring_int								
M3aadRQ36	Kōura	abun-	vs_ref-	af-	(Intercept)	0.307	0.533	0.575	0.565	NA	-	1.352	
		dance	er-	ter							0.739		
			ence										

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q	species	response	com- pari- type	time_frame	es- ti- mate	std.er- ror	stat- tic	p.value	adjco- B	Flow	low	high
M3aa	RQ3K	Kōura	abun-	vs_ref-af-	site	Stone_pile	- 0.923	- 0.805	NA	- 1.582			
			dance	er-ter			0.228	0.247		2.038			
				ence									
M3aa	RQ3K	Kōura	abun-	vs_ref-af-	monitoring_int		- 0.072	- 0.101	NA	- 0.023			
			dance	er-ter			0.118	1.640		0.260			
				ence									
M3aa	RQ3K	Kōura	abun-	vs_ref-af-	site-		0.100	0.131	0.760	0.447	0.447	- 0.358	
			dance	er-ter	Stone_pile:mon-						0.158		
				ence	itoring_int								
M3ab	RQ3K	Kōura	biomass	vs_ref-af-	(Intercept)		3.445	0.539	6.393	0.000	NA	2.389	4.502
				er-ter									
				ence									
M3ab	RQ3K	Kōura	biomass	vs_ref-af-	site	Stone_pile	- 0.934	- 0.488	NA	- 1.182			
				er-ter			0.648	0.694		2.477			
				ence									
M3ab	RQ3K	Kōura	biomass	vs_ref-af-	monitoring_int		- 0.097	- 0.178	NA	- 0.060			
				er-ter			0.131	1.346		0.322			
				ence									
M3ab	RQ3K	Kōura	biomass	vs_ref-af-	site-		0.231	0.173	1.338	0.181	0.243	- 0.569	
				er-ter	Stone_pile:mon-						0.107		
				ence	itoring_int								
M3ba	RQ3K	Cat-	abun-	ref-af-	(Intercept)		- 1.788	- 0.000	NA	- -			
		fish	dance	er-ter			6.715	3.757		10.219	3.212		
				ence									
M3ba	RQ3K	Cat-	abun-	ref-af-	Site_type_de-		0.731	1.033	0.707	0.479	NA	- 2.756	
		fish	dance	er-ter	tailReference_2						1.295		
				ence									

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q	species	response	com- pari- type	time_frame	es- ti- mate	std.er-	stat- tic	p.value	adjco- B	Flow	low	high
M3ba	RQ3K	Kōura	abun-	ref-	af-	Site_type_de-	-	0.567	-	0.642	NA	-	0.848
			dance	er-	ter	tailReference_2	0.263	0.464			1.374		
				ence									
M3ba	RQ3K	Kōura	abun-	ref-	af-	monitoring_int	-	0.075	-	0.069	NA	-	0.010
			dance	er-	ter		0.136	1.820			0.283		
				ence									
M3ba	RQ3K	Kōura	abun-	ref-	af-	SeasonSpring	-	0.210	-	0.103	NA	-	0.070
			dance	er-	ter		0.343	1.628			0.755		
				ence									
M3ba	RQ3K	Kōura	abun-	ref-	af-	SeasonSummer	-	0.225	-	0.349	NA	-	0.231
			dance	er-	ter		0.211	0.936			0.652		
				ence									
M3ba	RQ3K	Kōura	abun-	ref-	af-	SeasonWinter	-	0.242	-	0.000	NA	-	-
			dance	er-	ter		0.942	3.889			1.417	0.467	
				ence									
M3ba	RQ3K	Kōura	abun-	ref-	af-	Site_type_de-	-	0.101	-	0.802	0.902	-	0.173
			dance	er-	ter	tailRefer-	0.025	0.251			0.223		
				ence		ence_2:moni-							
						toring_int							
M3bb	RQ3K	Kōura	biomass	ref-	af-	(Intercept)	3.370	0.860	3.917	0.000	NA	1.683	5.056
				er-	ter								
				ence									
M3bb	RQ3K	Kōura	biomass	ref-	af-	Site_type_de-	-	0.737	-	0.803	NA	-	1.261
				er-	ter	tailReference_2	0.184	0.250			1.630		
				ence									
M3bb	RQ3K	Kōura	biomass	ref-	af-	monitoring_int	-	0.117	-	0.373	NA	-	0.125
				er-	ter		0.104	0.891			0.334		
				ence									

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1	Q3	K _{our}	biomass	ref- er- ence	time_frame	Site_type_de- tailRefer- ence_2:moni- toring_int	es- ti- mate	std.er- ror	stat- tic	p.value	adjco- BFlow	low	high
M3bb	RQ3K					af- ter		-	0.167	-	0.812	0.902	-	0.288
								0.040		0.237			0.368	

¹¹ Source: [Article Notebook](#)

Table 3: DHARMa goodness-of-fit tests per model. $p < 0.05$ indicates a significant departure from model assumptions (uniformity of residuals, over/under-dispersion, or excess outliers) and warrants closer inspection of the corresponding diagnostic plot above.

Model	Uniformity_p	Dispersion_p	Outlier_p
M2aac	0.139	0.952	0.11
M2abc	0.004	0.482	1.00
M2aoc	0.289	0.770	1.00
M2aak	0.708	0.906	1.00
M2abk	0.886	0.816	1.00
M2aok	0.331	0.952	1.00
M2bac	0.024	0.776	1.00
M2bbc	0.109	0.596	1.00
M2boc	0.084	0.638	1.00
M2bak	0.796	0.740	1.00
M2bbk	0.505	0.704	1.00
M2bok	0.468	0.896	1.00
M3aac	0.860	0.606	1.00
M3abc	0.551	0.346	1.00
M3aak	0.618	0.910	1.00
M3abk	0.994	0.724	1.00

Table 3: DHARMA goodness-of-fit tests per model. $p < 0.05$ indicates a significant departure from model assumptions (uniformity of residuals, over/under-dispersion, or excess outliers) and warrants closer inspection of the corresponding diagnostic plot above.

Model	Uniformity_p	Dispersion_p	Outlier_p
M3bbc	0.885	0.748	1.00
M3bak	0.959	0.878	1.00
M3bbk	0.872	0.896	1.00

12 Source: [Article Notebook](#)

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Figure 1: Map of the monitoring sites.

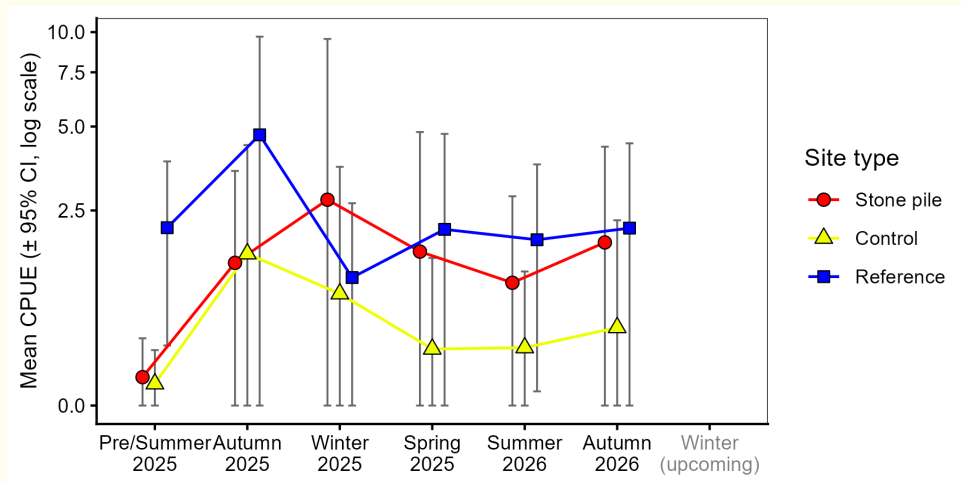


Figure 2: Mean catch per monitoring round ($\pm$ 95% CI) by site type for kōura.

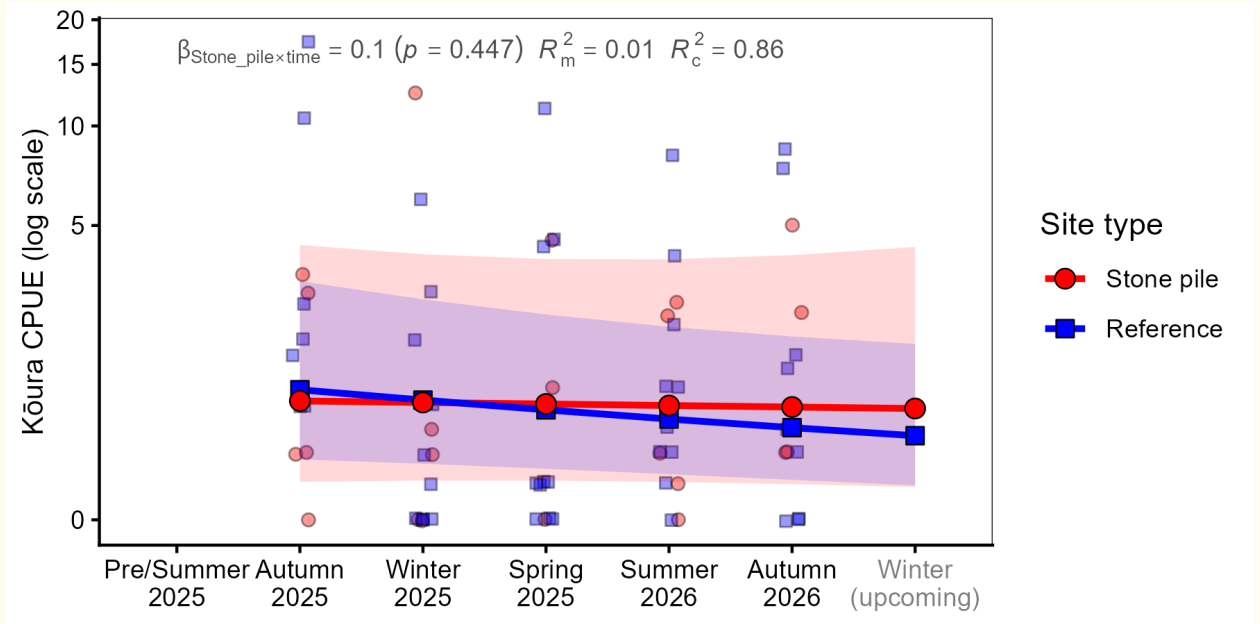


Figure 3: Predicted kōura CPUE trajectories over monitoring rounds at stone pile and reference sites from the RQ3 abundance convergence model. Shaded ribbons represent 95% confidence intervals. The annotation shows the key interaction term ($\pm$ p-value) and model R^2 values for fixed effects (R_m^2) and conditional (R_c^2).

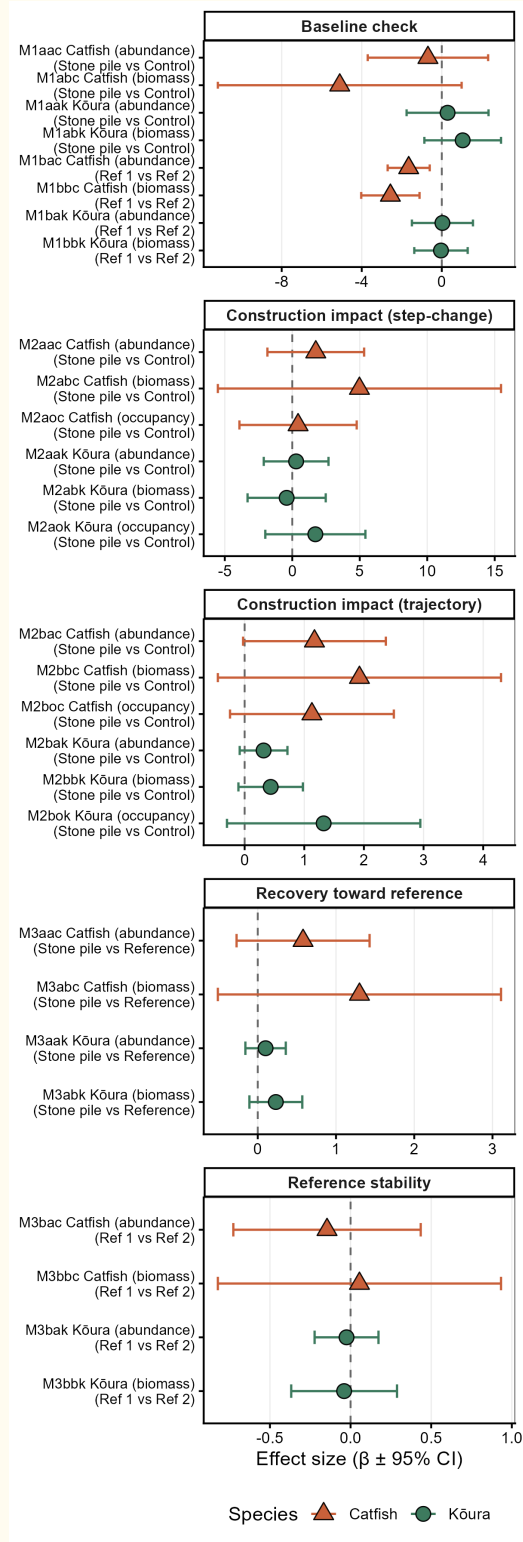


Figure 4: Effect sizes ($\beta \pm 95\%$ CI) for the key term of interest of every model in model_registry (kōura and catfish, RQ1–RQ4). Filled points indicate significant effects ($p < 0.05$). Each panel has its own x-axis scale— magnitudes are not directly comparable across panels because each RQ tests a different contrast (step-change, per-round slope, or simple group difference) and RQ1 occupancy is on the log-odds scale while all others use the Tweedie log link. Models with degenerate confidence intervals (non-positive-definite Hessian) are excluded.

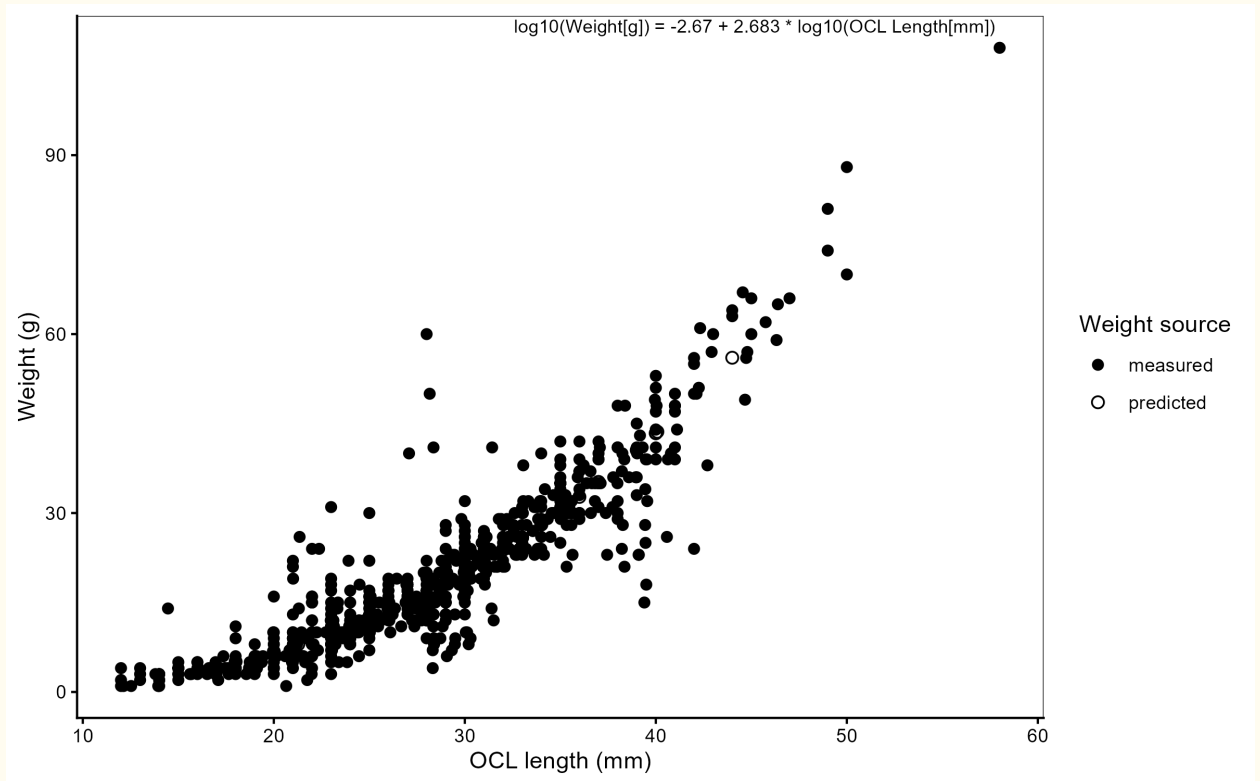


Figure 5: Length–weight relationship for kōura, fitted on log –log scale. Open points are weights predicted from the length–weight regression (used to fill missing scale measurements); filled points are directly measured weights.

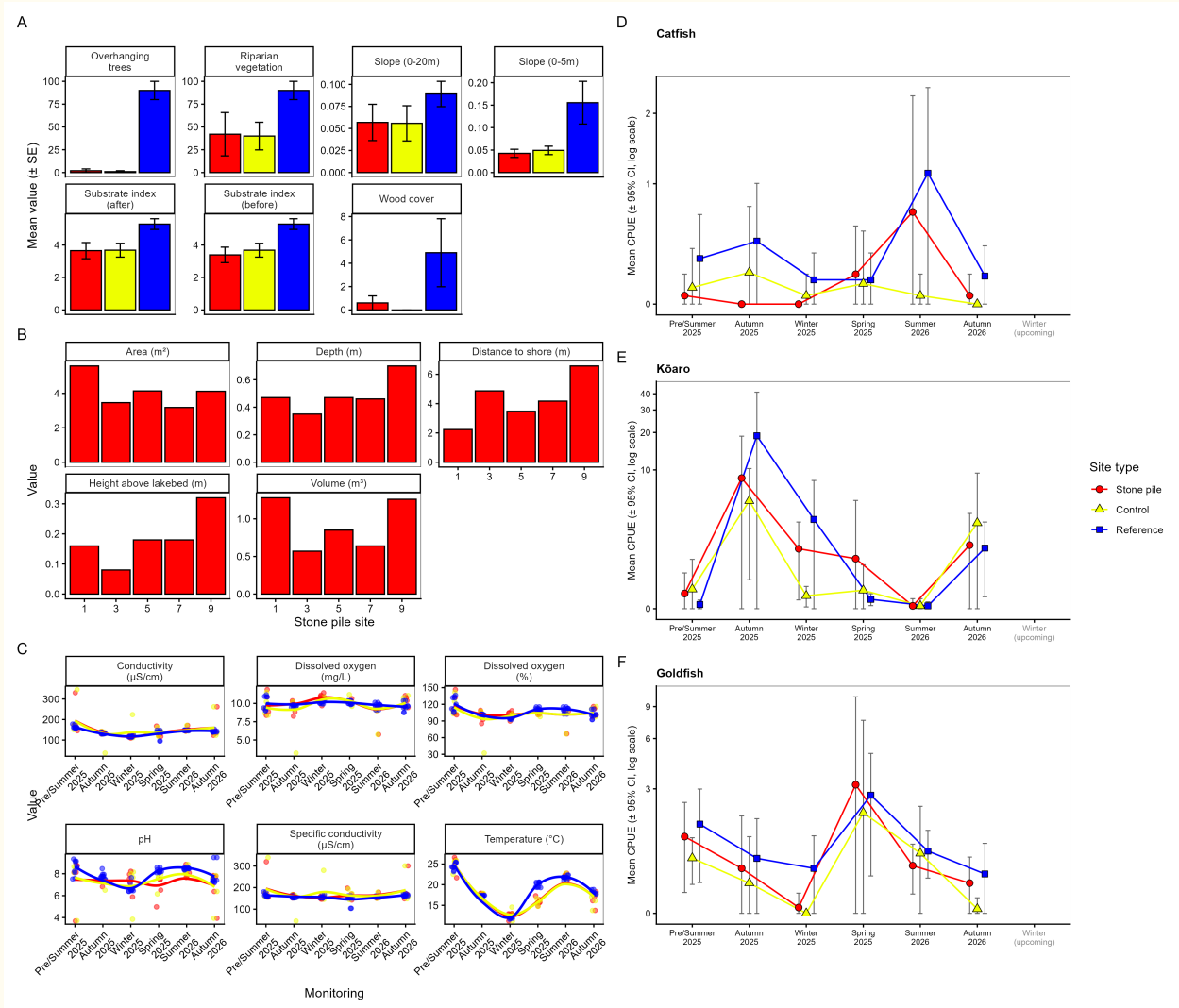


Figure 6: Descriptive context plots, combined. Left column (top to bottom): baseline habitat characteristics (mean $\pm$ SE by site type), stone pile structure dimensions (5 stone pile sites), chemical conditions over time (mean $\pm$ smoothed trend by site type). Right column: mean catch per monitoring round by site type, for catfish, kōaro, and goldfish. Site-type colours are shared across all panels — see the legend on the right. Background/methods-supporting figures, not primary results — see Results for the BACI, trend, and convergence models.

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